Chapter 2. Topological Spaces and Continuous Functions

Bases, order, product, subspace, metric and quotient topologies; closed sets; continuity

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2.1 Topological spaces

Motivation. A topology keeps exactly the properties of open sets in $\mathbb{R}^n$ that continuity and convergence use.

Definition (Topology)

A **topology** on a set X is a collection $\mathcal{T}$ of subsets of X, called **open sets**, such that $\emptyset, X \in \mathcal{T}$, arbitrary unions of members of $\mathcal{T}$ are in $\mathcal{T}$, and finite intersections of members of $\mathcal{T}$ are in $\mathcal{T}$.

Examples: the **discrete topology** $\mathcal{P}(X)$; the **trivial topology** $\{\emptyset, X\}$; the **finite complement topology** $\mathcal{T}_f = \{U : X - U \text{ finite}\} \cup \{\emptyset\}$ (a topology because $X - \bigcup U_\alpha = \bigcap (X - U_\alpha)$ and $X - \bigcap_{i \le n} U_i = \bigcup_{i \le n} (X - U_i)$ are finite); the countable complement topology. If $\mathcal{T} \subset \mathcal{T}'$ then $\mathcal{T}'$ is **finer** and $\mathcal{T}$ **coarser**.

2.2 Basis for a topology

Motivation. A basis, like the open discs in $\mathbb{R}^2$, generates all open sets, and most verifications need only be done on basis elements.

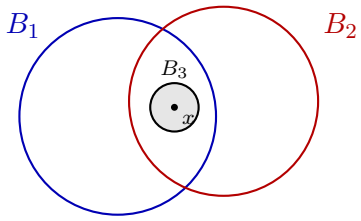


Figure. *The second basis axiom: if $x \in B_1 \cap B_2$, some basis element B_3 with $x \in B_3 \subset B_1 \cap B_2$ exists. This is what makes finite intersections of open sets open.*

Definition (Basis)

A collection $\mathcal{B}$ of subsets of X is a **basis** if (1) each x lies in at least one $B \in \mathcal{B}$, and (2) if $x \in B_1 \cap B_2$ then some $B_3 \in \mathcal{B}$ has $x \in B_3 \subset B_1 \cap B_2$. The topology **generated** by $\mathcal{B}$: U is open iff for each $x \in U$ there is $B \in \mathcal{B}$ with $x \in B \subset U$.

That this is a topology: unions are clear; for $U_1 \cap U_2$ and x in it, pick $B_i \ni x$ inside U_i and then $B_3 \subset B_1 \cap B_2$; finite intersections follow by induction. Examples: circular regions or rectangles in $\mathbb{R}^2$; one-point sets (discrete topology).

Lemma 2.1 (13.1)

If $\mathcal{B}$ is a basis for $\mathcal{T}$, then $\mathcal{T}$ equals the collection of all unions of elements of $\mathcal{B}$.

Proof. Unions of basis elements are open. An open U is the union of the basis elements B_x with $x \in B_x \subset U$, $x \in U$. ∎

Lemma 2.2 (13.2)

Let $\mathcal{C}$ be a collection of open sets of X such that for each open U and $x \in U$ there is $C \in \mathcal{C}$ with $x \in C \subset U$. Then $\mathcal{C}$ is a basis for the topology of X.

Proof. Condition (1) holds with $U = X$; for (2) apply the hypothesis to the open set $C_1 \cap C_2$. The topology $\mathcal{T}'$ generated by $\mathcal{C}$ contains $\mathcal{T}$ by hypothesis, and is contained in $\mathcal{T}$ because its sets are unions of members of $\mathcal{C} \subset \mathcal{T}$. ■

Lemma 2.3 (13.3)

Let $\mathcal{B}, \mathcal{B}'$ be bases for $\mathcal{T}, \mathcal{T}'$ on X. Then $\mathcal{T}'$ is finer than $\mathcal{T}$ iff for each x and each $B \in \mathcal{B}$ containing x there is $B' \in \mathcal{B}'$ with $x \in B' \subset B$.

Proof. ($\Leftarrow$) If $U \in \mathcal{T}$ and $x \in U$, choose B with $x \in B \subset U$, then B'; so $U \in \mathcal{T}'$.

($\Rightarrow$) $B \in \mathcal{T} \subset \mathcal{T}'$, and $\mathcal{T}'$ is generated by $\mathcal{B}'$. ■

On $\mathbb{R}$: the **standard topology** (basis: open intervals); the **lower limit topology** $\mathbb{R}_\ell$ (basis: $[a, b)$); the **K-topology** $\mathbb{R}_K$ (basis: (a, b) and $(a, b) - K$, $K = \{1/n : n \in \mathbb{Z}_+\}$).

Lemma 2.4 (13.4)

The topologies of $\mathbb{R}_\ell$ and $\mathbb{R}_K$ are strictly finer than the standard topology, but not comparable with one another.

Proof. Given $(a, b) \ni x$, both $[x, b)$ and (a, b) lie inside it, so both are finer (Lemma 2.3). $[x, d)$ contains no open interval around x, and $(-1, 1) - K$ contains no open interval around 0, so both are strictly finer. No basis element of $\mathbb{R}_K$ containing 10 lies in $[10, 11)$, and no basis element of $\mathbb{R}_\ell$ containing 0 lies in $(-1, 1) - K$. ■

A **subbasis** $\mathcal{S}$ is a collection of subsets whose union is X; the topology it generates is the collection of unions of finite intersections of members of $\mathcal{S}$ (these finite intersections form a basis).

2.3 The order topology

Motivation. Every linearly ordered set carries a topology generated by open intervals, giving test examples such as the ordered square.

Let X be simply ordered with more than one element. The **order topology** has as basis: all open intervals (a, b); all $[a_0, b)$ if X has a smallest element a_0; all $(a, b_0]$ if X has a largest element b_0. (If $x \in B_1 \cap B_2$, the intersection of two such sets is again of this form.) The open rays $(a, +\infty)$, $(-\infty, a)$ form a subbasis. Examples: $\mathbb{R}$ (standard topology); $\mathbb{Z}_+$ (discrete); $\mathbb{R} \times \mathbb{R}$ in the dictionary order (basis: vertical intervals $(a \times b, a \times d)$); $\{1, 2\} \times \mathbb{Z}_+$ in the dictionary order, which is not discrete since every neighbourhood of 2×1 contains points $1 \times n$.

2.4 The product topology on $X \times Y$

Motivation. Products of open sets form a basis for a topology on $X \times Y$.

The **product topology** on $X \times Y$ has basis $\mathcal{B} = \{U \times V : U \text{ open in } X,\ V \text{ open in } Y\}$ (a basis since $(U_1 \times V_1) \cap (U_2 \times V_2) = (U_1 \cap U_2) \times (V_1 \cap V_2)$).

Theorem 2.5 (15.1)

If $\mathcal{B}$ and $\mathcal{C}$ are bases for X and Y, then $\{B \times C : B \in \mathcal{B}, C \in \mathcal{C}\}$ is a basis for the topology of $X \times Y$.

Proof. Given open W and $x \times y \in W$, choose $U \times V$ with $x \times y \in U \times V \subset W$, then $B \subset U$, $C \subset V$ basis elements containing x, y. Apply Lemma 2.2. ■

Theorem 2.6 (15.2)

$\mathcal{S} = \{\pi_1^{-1}(U) : U \text{ open in } X\} \cup \{\pi_2^{-1}(V) : V \text{ open in } Y\}$ is a subbasis for the product topology.

Proof. $\pi_1^{-1}(U) \cap \pi_2^{-1}(V) = U \times V$, so the finite intersections of members of $\mathcal{S}$ are exactly the basis sets $U \times V$ (together with sets of the same form). ■

2.5 The subspace topology

Motivation. A subset inherits a topology from its surroundings; "open" then depends on the ambient space.

For $Y \subset X$, the **subspace topology** is $\mathcal{T}_Y = \{Y \cap U : U \in \mathcal{T}\}$.

Lemma 2.7 (16.1)

If $\mathcal{B}$ is a basis for X, then $\{B \cap Y : B \in \mathcal{B}\}$ is a basis for the subspace topology on Y.

Proof. For U open in X and $y \in U \cap Y$, choose B with $y \in B \subset U$; then $y \in B \cap Y \subset U \cap Y$. Apply Lemma 2.2. ■

Lemma 2.8 (16.2)

If U is open in Y and Y is open in X, then U is open in X.

Proof. $U = Y \cap V$ with V open in X, an intersection of two open sets of X. ■

Theorem 2.9 (16.3)

If A is a subspace of X and B of Y, the product topology on $A \times B$ equals the subspace topology from $X \times Y$.

Proof. $(U \times V) \cap (A \times B) = (U \cap A) \times (V \cap B)$; the left sides form a basis for the subspace topology (Lemma 2.7) and the right sides a basis for the product topology (Theorem 2.5). ■

Theorem 2.10 (16.4)

If X is ordered with the order topology and $Y \subset X$ is convex (for $a < b$ in Y, $(a, b) \subset Y$), then the order topology on Y equals the subspace topology.

Proof. The intersection of an open ray $(a, +\infty)$ of X with Y is Y, $\emptyset$, or an open ray of Y (convexity is used when $a \notin Y$: then a is a lower or upper bound of Y). So subbasis elements of the subspace topology are open in the order topology. Conversely, an open ray of Y is the intersection of Y with the corresponding ray of X. ■

Without convexity the two can differ: in $Y = [0, 1) \cup \{2\} \subset \mathbb{R}$, $\{2\}$ is open in the subspace topology but not in the order topology; in $I \times I$ with the dictionary order, the order topology differs from the subspace topology of $\mathbb{R} \times \mathbb{R}$ (dictionary order). We always give subsets of ordered sets the subspace topology unless said otherwise.

2.6 Closed sets and limit points

Motivation. Closures and limit points describe approximation; the Hausdorff condition restores uniqueness of limits.

A is **closed** if $X - A$ is open.

Theorem 2.11 (17.1)

$\emptyset$ and X are closed; arbitrary intersections and finite unions of closed sets are closed.

Proof. De Morgan's laws applied to the axioms for open sets. ■

Theorem 2.12 (17.2)

For a subspace Y, A is closed in Y iff $A = C \cap Y$ for some closed $C \subset X$.

Proof. If $A = C \cap Y$, then $Y - A = (X - C) \cap Y$ is open in Y. Conversely, $Y - A = U \cap Y$ with U open, and $A = (X - U) \cap Y$. ■

Theorem 2.13 (17.3)

If A is closed in Y and Y is closed in X, then A is closed in X.

The **interior** $\operatorname{Int} A$ is the union of the open sets contained in A; the **closure** $\overline{A}$ is the intersection of the closed sets containing A. So $\operatorname{Int} A \subset A \subset \overline{A}$, A is open iff $A = \operatorname{Int} A$, closed iff $A = \overline{A}$.

Theorem 2.14 (17.4)

If Y is a subspace of X, $A \subset Y$, and $\overline{A}$ is the closure of A in X, then the closure of A in Y is $\overline{A} \cap Y$.

Proof. $\overline{A} \cap Y$ is closed in Y and contains A, so it contains the closure B of A in Y. Conversely $B = C \cap Y$ with C closed in X (Theorem 2.12) and $C \supset A$, so $\overline{A} \subset C$ and $\overline{A} \cap Y \subset B$. ■

We say U is a **neighbourhood** of x if U is open and contains x, and that U **intersects** A if $U \cap A \neq \emptyset$.

Theorem 2.15 (17.5)

(a) $x \in \overline{A}$ iff every open set containing x intersects A.
(b) If the topology is given by a basis, $x \in \overline{A}$ iff every basis element containing x intersects A.

Proof. (a) Contrapositive: $x \notin \overline{A}$ iff some open $U \ni x$ misses A (take $U = X - \overline{A}$; conversely $X - U$ is a closed set containing A and missing x).

(b) Every open set containing x contains a basis element containing x. ■

For example, in $\mathbb{R}$ the closure of $(0, 1]$ is $[0, 1]$, of $\mathbb{Q}$ is $\mathbb{R}$, of $\{1/n\}$ is $\{0\} \cup \{1/n\}$. The point x is a **limit point** of A if every neighbourhood of x intersects A in a point other than x; equivalently $x \in \overline{A - \{x\}}$. Let A' denote the set of limit points.

Theorem 2.16 (17.6)

$\overline{A} = A \cup A'$.

Proof. $A' \subset \overline{A}$ by Theorem 2.15. If $x \in \overline{A} - A$, every neighbourhood of x meets A in a point, which is not x; so $x \in A'$. ■

Corollary 2.17 (17.7)

A subset is closed iff it contains all its limit points.

A sequence (x_n) **converges** to x if for every neighbourhood U of x there is N with $x_n \in U$ for $n \geq N$. In general limits need not be unique. X is a **Hausdorff space** if distinct points have disjoint neighbourhoods.

Theorem 2.18 (17.8)

Every finite point set in a Hausdorff space is closed.

Proof. For a single point x_0: any $x \neq x_0$ has a neighbourhood missing a neighbourhood of x_0, in particular missing x_0, so $x \notin \overline{\{x_0\}}$. Finite unions of closed sets are closed. ■

The condition that finite sets are closed is the T_1 **axiom**; it is weaker than Hausdorff (finite complement topology on $\mathbb{R}$).

Theorem 2.19 (17.9)

In a T_1 space, x is a limit point of A iff every neighbourhood of x contains infinitely many points of A.

Proof. ($\Rightarrow$) If a neighbourhood U of x met A in finitely many points, and $x_1, \ldots, x_m$ are those different from x, then $U - \{x_1, \ldots, x_m\}$ is a neighbourhood of x (finite sets are closed) meeting $A - \{x\}$ not at all.

($\Leftarrow$) is clear. ■

Theorem 2.20 (17.10)

In a Hausdorff space a sequence converges to at most one point.

Proof. If $x_n \to x$ and $y \neq x$, take disjoint neighbourhoods $U \ni x$, $V \ni y$; U contains all but finitely many x_n, so V cannot, and $x_n \not\to y$. ■

Theorem 2.21 (17.11)

Every simply ordered set is Hausdorff in the order topology; products of two Hausdorff spaces and subspaces of Hausdorff spaces are Hausdorff.

Proof. For $a < b$ in an ordered set: if some c lies between them, $(-\infty, c)$ and $(c, +\infty)$ separate; otherwise $(-\infty, b)$ and $(a, +\infty)$ do. For products use $U_1 \times V_1$ and $U_2 \times V_2$ (separating in one coordinate suffices); for subspaces intersect with the subspace. ■

2.7 Continuous functions

Motivation. Continuity means preimages of open sets are open; below are the standard ways to build continuous maps.

$f : X \to Y$ is **continuous** if $f^{-1}(V)$ is open for every open $V \subset Y$; it suffices to check V in a basis or subbasis, since preimages preserve unions and intersections.

Theorem 2.22 (18.1)

For $f : X \to Y$ the following are equivalent:
(1) f is continuous;
(2) $f(\overline{A}) \subset \overline{f(A)}$ for every $A \subset X$;
(3) $f^{-1}(B)$ is closed for every closed $B \subset Y$;
(4) for each x and each neighbourhood V of $f(x)$ there is a neighbourhood U of x with $f(U) \subset V$.

Proof. (1)⇒(2): for $x \in \overline{A}$ and a neighbourhood V of $f(x)$, $f^{-1}(V)$ is a neighbourhood of x meeting A in some y, and $f(y) \in V \cap f(A)$. (2)⇒(3): with $A = f^{-1}(B)$, $f(\overline{A}) \subset \overline{f(A)} \subset \overline{B} = B$, so $\overline{A} \subset A$. (3)⇒(1): complements. (1)⇒(4): $U = f^{-1}(V)$. (4)⇒(1): $f^{-1}(V)$ is the union of the neighbourhoods U_x given by (4). ■

A **homeomorphism** is a continuous bijection with continuous inverse; equivalently a bijection with $f(U)$ open iff U open. A continuous injection $f : X \to Y$ that is a homeomorphism onto $f(X)$ (subspace topology) is an **imbedding**. Examples: the identity $\mathbb{R}_\ell \to \mathbb{R}$ is a continuous bijection but not a homeomorphism; $[0, 1) \to S^1$, $t \mapsto (\cos 2\pi t, \sin 2\pi t)$ is a continuous bijection that is not a homeomorphism.

Theorem 2.23 (18.2, rules for constructing continuous functions)

(a) Constant functions are continuous.
(b) Inclusions of subspaces are continuous.
(c) Composites of continuous functions are continuous.
(d) Restricting the domain of a continuous function to a subspace keeps it continuous.
(e) Restricting or expanding the range: if $f : X \to Y$ is continuous and $Z \subset Y$ contains $f(X)$, then $f : X \to Z$ is continuous; if $Y \subset Z$ is a subspace, $f : X \to Z$ is continuous.
(f) Local formulation: f is continuous if X is a union of open sets U_α with each $f|_{U_\alpha}$ continuous.

Proof. (a) Preimages are $\emptyset$ or X.

(b) $j^{-1}(U) = U \cap A$.

(c) $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$.

(d) Composite with the inclusion.

(e) $f^{-1}(Z \cap V) = f^{-1}(V)$; and composite with the inclusion $Y \to Z$.

(f) $f^{-1}(V) = \bigcup (f|_{U_\alpha})^{-1}(V)$, a union of sets open in U_α, hence open in X (Lemma 2.8). ■

Theorem 2.24 (18.3, the pasting lemma)

Let $X = A \cup B$ with A, B closed, $f : A \to Y$ and $g : B \to Y$ continuous and $f = g$ on $A \cap B$. Then $h = f \cup g : X \to Y$ is continuous. (The same holds with A, B open.)

Proof. For closed $C \subset Y$, $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$; each is closed in A or B, hence in X (Theorem 2.13). ∎

Theorem 2.25 (18.4, maps into products)

$f : A \to X \times Y$, $f = (f_1, f_2)$, is continuous iff f_1 and f_2 are continuous.

Proof. The projections are continuous ($\pi_1^{-1}(U) = U \times Y$), so $f_i = \pi_i \circ f$ are continuous if f is. Conversely $f^{-1}(U \times V) = f_1^{-1}(U) \cap f_2^{-1}(V)$. ∎

2.8 The product topology

Motivation. For infinite products the product topology, not the box topology, makes a map continuous exactly when each coordinate is.

For an indexed family $\{X_\alpha\}_{\alpha \in J}$, the **box topology** on $\prod X_\alpha$ has basis all products $\prod U_\alpha$ of open sets, and the **product topology** has subbasis $\mathcal{S} = \bigcup_\beta \{\pi_\beta^{-1}(U_\beta)\}$.

Theorem 2.26 (19.1, comparison)

The box topology has basis all $\prod U_\alpha$ with U_α open; the product topology has basis all $\prod U_\alpha$ with U_α open and $U_\alpha = X_\alpha$ for all but finitely many α. For finite products they coincide; in general the box topology is finer.

Proof. A finite intersection $\pi_{\beta_1}^{-1}(U_{\beta_1}) \cap \cdots \cap \pi_{\beta_n}^{-1}(U_{\beta_n})$ is exactly a product $\prod U_\alpha$ with $U_\alpha = X_\alpha$ except for $\alpha \in \{\beta_1, \ldots, \beta_n\}$. ∎

Theorem 2.27 (19.2)

If each X_α has a basis $\mathcal{B}_\alpha$, products $\prod B_\alpha$ with $B_\alpha \in \mathcal{B}_\alpha$ form a basis for the box topology, and those with $B_\alpha \in \mathcal{B}_\alpha$ for finitely many α and $B_\alpha = X_\alpha$ otherwise form a basis for the product topology.

Theorem 2.28 (19.3)

If $A_\alpha \subset X_\alpha$, then $\prod A_\alpha$ is a subspace of $\prod X_\alpha$ when both carry the box topology, or both the product topology.

Theorem 2.29 (19.4)

A product of Hausdorff spaces is Hausdorff in both topologies.

Theorem 2.30 (19.5)

In either topology, $\prod \overline{A_\alpha} = \overline{\prod A_\alpha}$.

Proof. If $x \in \prod \overline{A_\alpha}$ and $U = \prod U_\alpha$ is a basis element containing x, choose $y_\alpha \in U_\alpha \cap A_\alpha$; then $y \in U \cap \prod A_\alpha$. Conversely, if $x \in \overline{\prod A_\alpha}$ and $V_\beta \ni x_\beta$ is open, $\pi_\beta^{-1}(V_\beta)$ meets $\prod A_\alpha$, so V_β meets A_β. (Theorems 2.27–2.29 are proved like their two-factor versions.) ∎

Theorem 2.31 (19.6)

Let $f : A \to \prod X_\alpha$, $f(a) = (f_\alpha(a))$, where $\prod X_\alpha$ has the product topology. Then f is continuous iff each f_α is continuous.

Proof. π_β is continuous, giving necessity. Conversely, $f^{-1}(\pi_\beta^{-1}(U_\beta)) = f_\beta^{-1}(U_\beta)$ is open, and these sets form a subbasis. ■

This fails for the box topology: $f : \mathbb{R} \to \mathbb{R}^\omega$, $f(t) = (t, t, \dots)$, has continuous coordinates, but $f^{-1}(\prod(-\frac{1}{n}, \frac{1}{n})) = \{0\}$. Henceforth products carry the product topology.

2.9 The metric topology

Motivation. The standard metrics on $\mathbb{R}^n$ give the product topology, and the uniform metric on $\mathbb{R}^J$ lies strictly between the product and box topologies.

A **metric** d on X satisfies $d(x, y) \geq 0$ with equality iff $x = y$, symmetry, and the triangle inequality. The ϵ-balls $B_d(x, \epsilon)$ form a basis (if $y \in B_d(x, \epsilon)$, then $B_d(y, \epsilon - d(x, y)) \subset B_d(x, \epsilon)$), generating the **metric topology**; U is open iff each $y \in U$ has a ball $B_d(y, \delta) \subset U$. X is **metrizable** if some metric induces its topology. A is **bounded** if $\sup_{a,b\in A} d(a, b) = \operatorname{diam} A < \infty$.

Theorem 2.32 (20.1)

$\bar{d}(x, y) = \min\{d(x, y), 1\}$ is a metric inducing the same topology as d (the **standard bounded metric**).

Proof. Triangle inequality: if $d(x, y) \geq 1$ or $d(y, z) \geq 1$ the right side is at least 1; otherwise $\bar{d}(x, z) \leq d(x, z) \leq d(x, y) + d(y, z) = \bar{d}(x, y) + \bar{d}(y, z)$. The balls of radius < 1 coincide and form a basis for both topologies. ■

Lemma 2.33 (20.2)

For metrics d, d' on X with topologies $\mathcal{T}, \mathcal{T}'$, $\mathcal{T}'$ is finer than $\mathcal{T}$ iff for each x and $\epsilon > 0$ there is $\delta > 0$ with $B_{d'}(x, \delta) \subset B_d(x, \epsilon)$.

Proof. This is Lemma 2.3 for the bases of balls, using that every ball containing x contains a ball centred at x. ■

Theorem 2.34 (20.3)

The euclidean metric d and the square metric $\rho(x, y) = \max|x_i - y_i|$ on $\mathbb{R}^n$ both induce the product topology.

Proof. $\rho \leq d \leq \sqrt{n}\rho$ gives $B_d(x, \epsilon) \subset B_\rho(x, \epsilon)$ and $B_\rho(x, \epsilon/\sqrt{n}) \subset B_d(x, \epsilon)$, so they agree. Product basis elements $\prod(a_i, b_i) \ni x$ contain $B_\rho(x, \epsilon)$ for $\epsilon = \min_i \min(x_i - a_i, b_i - x_i)$, and $B_\rho(x, \epsilon)$ is itself a product of intervals. ■

On $\mathbb{R}^J$ the **uniform metric** is $\bar{\rho}(x, y) = \sup_\alpha \bar{d}(x_\alpha, y_\alpha)$.

Theorem 2.35 (20.4)

The uniform topology on $\mathbb{R}^J$ is finer than the product topology and coarser than the box topology; if J is infinite, all three are different.

Proof. A product basis element $\prod U_\alpha$ restricting $\alpha_1, \ldots, \alpha_n$ contains a ball $B_{\bar{\rho}}(x, \epsilon)$ (ϵ the minimum of the radii needed in those coordinates, and < 1). A ball $B_{\bar{\rho}}(x, \epsilon)$ contains the box set $\prod(x_\alpha - \epsilon/2, x_\alpha + \epsilon/2)$. For infinite J: $B_{\bar{\rho}}(0, 1)$ is not product-open, and $\prod(-1/n, 1/n)$ (indexing a countable subset) contains no uniform ball around 0. ■

Theorem 2.36 (20.5)

With $\bar{d}(a, b) = \min\{|a - b|, 1\}$, $D(x, y) = \sup_i \bar{d}(x_i, y_i)/i$ is a metric inducing the product topology on $\mathbb{R}^\omega$.

Proof. D is a metric (sup of metrics divided by i). Given U open in the product topology and $x \in U$, choose a basis set $\prod_{i \le N}(x_i - \epsilon, x_i + \epsilon) \times \mathbb{R} \times \cdots \subset U$, $\epsilon < 1$; if $D(x, y) < \epsilon/N$ then $\bar{d}(x_i, y_i) < \epsilon$ for $i \le N$, so $B_D(x, \epsilon/N) \subset U$. Conversely, given $B_D(x, \epsilon)$, choose N with $1/N < \epsilon$ and let $V = \prod_{i<N}(x_i - \epsilon/2, x_i + \epsilon/2) \times \mathbb{R} \times \mathbb{R} \times \cdots$. For $y \in V$, $\bar{d}(x_i, y_i)/i \le \epsilon/2$ for $i < N$ and $\le 1/N$ for $i \ge N$, so $D(x, y) \le \max(\epsilon/2, 1/N) < \epsilon$ and $V \subset B_D(x, \epsilon)$. ■

2.10 The metric topology (continued)

Motivation. In metric spaces sequences detect closure and continuity, and uniform limits of continuous functions are continuous; both can fail in general spaces.

Subspaces of metric spaces are metric spaces (restrict d); order topologies may or may not be metrizable; Hausdorff is necessary for metrizability.

Theorem 2.37 (21.1)

For metrizable X, Y with metrics d_X, d_Y, $f : X \to Y$ is continuous iff for each x and $\epsilon > 0$ there is $\delta > 0$ with $d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \epsilon$.

Proof. Apply Theorem 2.22(4) with neighbourhoods taken to be balls. ■

Lemma 2.38 (21.2, the sequence lemma)

If a sequence of points of A converges to x, then $x \in \overline{A}$; the converse holds if X is metrizable (indeed if X has a countable neighbourhood basis at x).

Proof. Every neighbourhood of x contains points of the sequence. Conversely, if $x \in \overline{A}$ and X is metrizable, choose $x_n \in B(x, 1/n) \cap A$; then $x_n \to x$. ■

Theorem 2.39 (21.3)

If f is continuous, then $x_n \to x$ implies $f(x_n) \to f(x)$; the converse holds if X is metrizable.

Proof. If $V \ni f(x)$ is open, $f^{-1}(V) \ni x$ contains x_n for large n. Conversely, for metrizable X we show $f(\overline{A}) \subset \overline{f(A)}$: if $x \in \overline{A}$, choose $x_n \in A$ with $x_n \to x$ (Lemma 2.38); then $f(x_n) \to f(x)$, so $f(x) \in \overline{f(A)}$. Apply Theorem 2.22. ■

In particular $\mathbb{R}^\omega$ with the box topology and $\mathbb{R}^J$ (J uncountable) in the product topology are not metrizable: in both, a point lies in the closure of a set without being a limit of a sequence from it.

Lemma 2.40 (21.4)

Addition, subtraction and multiplication $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ and division $\mathbb{R} \times (\mathbb{R} - \{0\}) \to \mathbb{R}$ are continuous.

Proof. Standard ϵ–δ estimates using the metric $|a - a'| + |b - b'|$ on $\mathbb{R}^2$; e.g. $|ab - a'b'| \leq |a||b - b'| + |b'||a - a'|$. ■

Theorem 2.41 (21.5)

If $f, g : X \to \mathbb{R}$ are continuous, so are $f \pm g$, fg, and f/g where $g \neq 0$.

Proof. $f + g$ is the composite of $(f, g) : X \to \mathbb{R} \times \mathbb{R}$ (Theorem 2.25) and addition (Lemma 2.40). ■

$f_n : X \to Y$ (Y metric) **converges uniformly** to f if for every ϵ there is N with $d(f_n(x), f(x)) < \epsilon$ for all $n \geq N$ and all x.

Theorem 2.42 (21.6, uniform limit theorem)

If continuous $f_n : X \to Y$ converge uniformly to f, then f is continuous.

Proof. Let V be open in Y, $x_0 \in f^{-1}(V)$, and $B(f(x_0), \epsilon) \subset V$. Choose N with $d(f_N(x), f(x)) < \epsilon/3$ for all x, and a neighbourhood U of x_0 with $f_N(U) \subset B(f_N(x_0), \epsilon/3)$. For $x \in U$, $d(f(x), f(x_0)) < \epsilon$, so $f(U) \subset V$. ■

2.11 The quotient topology

Motivation. The quotient topology formalizes gluing, and its universal property decides continuity of maps on the glued space.

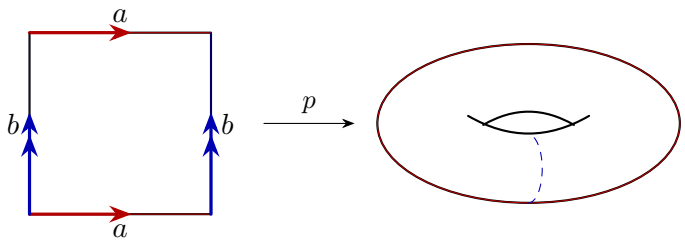


Figure. *Gluing opposite sides of a square (identify the edges labelled a, and those labelled b, respecting arrows) gives the torus; the quotient map p carries the quotient topology.*

A surjective map $p : X \to Y$ is a **quotient map** if $U \subset Y$ is open iff $p^{-1}(U)$ is open. Equivalently, p is continuous and maps **saturated** open sets (sets of the form $p^{-1}(V)$) to open sets. Open or closed continuous surjections are quotient maps (e.g. the projection $\pi_1 : X \times Y \to X$ is open). If X is a space and $p : X \to A$ a surjection onto a set, the **quotient topology** on A is the unique topology making p a quotient map; for a partition X^* of X we get the **quotient space** X^*.

Theorem 2.43 (22.1)

Let $p : X \to Y$ be a quotient map, $A \subset X$ saturated, and $q = p|_A : A \to p(A)$.
(1) If A is open or closed, q is a quotient map.
(2) If p is open or closed, q is a quotient map.

Proof. For $V \subset p(A)$, saturation gives $q^{-1}(V) = p^{-1}(V)$ and $p(U \cap A) = p(U) \cap p(A)$ for U open (or closed).

(1) If A is open and $q^{-1}(V)$ is open in A, it is open in X, so $p^{-1}(V)$ is open and V is open in Y, hence in $p(A)$.

(2) If p is open and $q^{-1}(V) = U \cap A$ with U open, then $V = q(U \cap A) = p(U) \cap p(A)$ is open in $p(A)$. Closed cases are the same with closed sets. ■

Composites of quotient maps are quotient maps; products of quotient maps need not be; quotients of Hausdorff spaces need not be Hausdorff.

Theorem 2.44 (22.2)

Let $p : X \to Y$ be a quotient map and $g : X \to Z$ constant on each fibre $p^{-1}(y)$. Then g induces $f : Y \to Z$ with $f \circ p = g$; f is continuous iff g is, and f is a quotient map iff g is.

Proof. f is well defined. For $V \subset Z$ open, $g^{-1}(V) = p^{-1}(f^{-1}(V))$ is open iff $f^{-1}(V)$ is open (quotient map). If g is a quotient map, f is surjective, and $f^{-1}(V)$ open implies $g^{-1}(V)$ open, hence V open; conversely composites of quotient maps are quotient maps. ■

Corollary 2.45 (22.3)

Let $g : X \to Z$ be a continuous surjection and $X^* = \{g^{-1}(z)\}$ with the quotient topology.
(a) g induces a continuous bijection $f : X^* \to Z$, which is a homeomorphism iff g is a quotient map.
(b) If Z is Hausdorff, so is X^*.

Proof. (a) is Theorem 2.44.

(b) Disjoint open sets in Z pull back by the continuous f to disjoint open sets in X^*. ■

Example 2.1

(a) Collapsing ∂B^2 to a point gives S^2; identifying opposite edges of the square gives the torus.
(b) $p : \mathbb{R} \to S^1$, $p(t) = (\cos 2\pi t, \sin 2\pi t)$ is a quotient map (it is open), so $\mathbb{R}/\mathbb{Z} \cong S^1$ by Corollary 2.45.
(c) For $\mathbb{Q} \subset \mathbb{R}$ collapsed to a point, $p \times \mathrm{id}_{\mathbb{Q}} : \mathbb{R} \times \mathbb{Q} \to \mathbb{R}^* \times \mathbb{Q}$ is not a quotient map (Munkres Example 22.7).